

Modelling cultural ecosystem services in agricultural dykelands and tidal wetlands to inform coastal infrastructure decisions: A social media data approach

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ABSTRACT

Agricultural dykelands and tidal wetlands around Canada's Bay of Fundy are experiencing increasingly severe impacts from sea level rise and climate change, leading to management challenges. Managers will have to decide which dykes to reinforce, which to realign or remove, and where to restore wetlands. These decisions will have important impacts on the ecosystem services provided by different landscapes and the beneficiaries who use them. Cultural ecosystem services (CES) are non-material benefits that play a significant role in human-nature relationships. Previous research in the region has indicated strong local dykeland attachments and CES but provided little insight about tidal wetland uses and values. Using the Kings County region as a study area, we identify CES provided by the area through text analysis of Instagram data and map the CES using the SolVES model combined with environmental data. Revealed differences in CES delivery from agricultural dykelands and tidal wetlands provide spatial insight for coastal planning. The results show that agricultural dykelands have a higher supply capacity for cultural heritage/diversity, education and knowledge systems and social relations and relational values; tidal wetlands have a higher supply capacity for sense of place and terror. For aesthetics, inspiration and art, and recreation and tourism, the two landscapes are comparable in delivery. In the small proportion of the two landscapes where CES are modelled as being present they often overlap, demonstrating multifunctionality, with an average of 1.4 services provided by a given area of agricultural dykelands and 2.6 services by tidal wetlands.

1. Introduction

The area of tidal wetlands in northeastern North America, including the Bay of Fundy, has been declining for centuries under the influence of human activities [1]. Tidal wetlands are dynamic ecosystems located in transition areas between land and sea [2], which are extremely ecologically important and visually striking [3]. When estimated in economic terms tidal wetlands are one of the most valuable ecosystems on Earth and provide multiple ecosystem services (ES) that contribute to human well-being [4,5], such as biodiversity maintenance [6], carbon sequestration [7], coastal protection [8], recreational value [9], water purification and habitat provision [10,11].

Agricultural dykelands (also called polders), are areas that were drained using dykes and aboiteaux (one-way drains or sluices) to convert tidal wetland areas to agricultural land [12,13]. This approach was used by French settlers (called Acadians) who came to the Bay of Fundy in the 17th century and Nova Scotia's dyked landscape remains closely associated with Acadian culture as recognized with UNESCO World Heritage status [14,15]. Today, dykelands continue to provide food and ensure coastal safety from flooding [12,16], bringing both economic benefits and rich cultural ecosystem services. For instance, hiking, bird watching, running and biking are common public uses on the dykes themselves [17,18].

Cultural ecosystem services (CES) are non-material benefits that play

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a significant role in human-nature relationships and human well-being [19,20]. The Millennium Ecosystem Assessment defines CES as a variety of intangible values and benefits provided by the ecosystem, i.e. the benefits obtained through spiritual enrichment, cognitive development, introspection, recreation, and aesthetic experience [21]. In recent years, CES provision from landscapes has been researched in depth [22,23], with a growing amount of empirical research recognizing that CES are an integral element of the decision-making process for sustainable landscape management [24–26]. CES are gradually being incorporated into policy-making and environmental management frameworks [27, 28].

Scholars use a variety of methods to understand the usage and delivery of CES, such as semi-structured interviews [29], participatory mapping [30], monetary valuations [31], geospatial models [32], and social media data [33,34]. Social media data reveals the opinions, personal feelings and daily activities of users, providing a useful data source for quantifying CES [35]. Though not without its own challenges, analyzing social media data can help overcome some of the challenges associated with traditional social science methods by being less labor-intensive, less time-consuming, providing larger sample sizes and addressing the problem of low youth engagement [36–38].

Social media data have been widely used to identify CES under various classification frameworks, but manual analysis approaches can be time consuming with large-sized datasets [39,40]. Most recently, machine learning models have shown great potential to automatically label and attach keywords to photos posted on social media, using clustering to associate keywords to CES categories [41,42]. An alternate approach focuses on the captions of photos, analyzing user-generated contents (e.g., captions and comments) for expressions of CES delivery and developing a classification that assigns tags to CES categories [43, 44]. There is already research that focuses on textual data from social media posts that appears without images. Mitchell et al. conducted an automated content analysis of 80 million Twitter text data to examine how happiness varies by region in the United States [45]. Becken et al. used geo-tagged Twitter posts to measure users' sentiments about nature [46]. However, researchers have not yet made full use of the information contained in textual data associated with social media photos [47], which can convey views and opinions that cannot be obtained from the image contents [48].

Sea level rise and climate change are creating challenges for the management of agricultural dykelands and tidal wetlands around the Bay of Fundy. Maintaining and reinforcing the approximately 240 km of dykes on the Nova Scotia side of the Bay is economically unaffordable and no longer technically feasible [17]. Managers must decide which dykes to reinforce, which to realign or remove, and where to restore dykelands back to wetlands. The Nova Scotia Department of Agriculture with responsibility for the dyke system currently has a 8-year capital campaign underway to modernize the dyke system by making and implementing such decisions [49]. CES-related information can support decision-making, as CES reflect the landscape-related concerns of local communities [50]. Yet doing CES work in sparsely populated rural areas is difficult: there are fewer beneficiaries than in cities and reaching a sample of them is harder [51]. Previous work has provided little insight about perceptions and use of tidal wetlands, in part because few people are familiar with them after their long extirpation from the area or call the landscapes by name on social media [17,52]. Geo-tagged social media data can help bridge this knowledge gap by providing a large amount of point-level data from its user base.

In this paper, the study area is a rural area around a tidal river adjacent to the Bay of Fundy. To model CES delivered by agricultural dykelands and tidal wetland landscapes using social media, we established three specific goals: (1) identifying the CES categories offered in the area through text analysis of Instagram data; (2) combining the geo-tagged point data with a geospatial model, SolVES, to map CES in the study area; and, (3) analyzing the likelihood of agricultural dykelands and tidal wetlands delivering different services. The results convey the

CES provision of each alternative landscape as experienced by Instagram users but also raise questions about researching non-urban land uses in such ways. At the same time, we contribute to the evidence base about policy associated with regional climate change and dyke realignment.

2. Material and methods

2.1. Study area

The study area is in Kings County, Nova Scotia, Canada, adjacent to the Minas Basin of the megatidal Bay of Fundy. The study area includes rivers such as the Habitant, Ji'juktuk'kwejk/Cornwallis, and Gaspereau (Fig. 1). It serves as a good case study area because it houses active agriculture and a rich local food culture with extensive suburbanization, as well as recreational and tourism demands on the dykeland landscape. For instance, the UNESCO-listed Landscape of Grand-Pré is in the study area and comprises the largest area of intact dykeland in the region, which continues to be farmed, while being an international tourist destination and ancestral home for the Acadian diaspora. Research on these landscapes can show how CES evidence from social media can be used to guide coastal infrastructure decisions, with implications for the coastal adaptation challenges faced in other places with dyked land, e.g. the UK, Netherlands, etc.

2.2. Data sources

2.2.1. Social media data collection

This study selected Instagram as the platform for obtaining data. Instagram is one of the world's most popular social media platforms [53, 54]. On the Instagram platform in 2022, the percentage of users aged 13–24 is 40.8%, 45.1% of users aged 25–44% and 14.1% of users aged 45 and above [55]. It is used for capturing, sharing and promoting a variety of topics, including culturally relevant events and documenting everyday life, with over 2 billion active users [38,56,57]. Netlytic is an API-based tool that provides access to multiple social media sites. In 2018, before Instagram closed their public API, we used Netlytic to collect Instagram posts, including images, captions, and the metadata, based on 44 geographic locations around the dykelands of the Nova Scotia side of the Bay of Fundy (Fig. 1). Instagram posts geotagged within a 5 km radius around each location were retrieved, covering an area of approximately 3000 square kilometers on the shore. Data collection was conducted from April 9th to September 9th (capturing winter, summer and fall conditions in this region); see Chen, Caese-maecker et al. (2020) for more information. The total number of Instagram posts retrieved was 123,749, but for the study area 28,540.

2.2.2. Environmental data

Environmental and socioeconomic data can help us understand the drivers of CES, although it has been observed that CES are less landscape-associated than other ES [58]. We gathered datasets that we hypothesized to be associated with or drivers of CES supply in the region. The environmental data required for simulation is composed of: Euclidean distance to roads, water bodies (lakes and rivers), vineyard, dykes, agricultural dykeland, tidal wetland and coast were calculated, together with population density, land use, landform and elevation (Table 1) [24,59]. The spatial resolution of each data layer was resampled to 20 m to match the available elevation data. ArcGIS 10.5 was used to process all relevant geospatial and environmental data in the study area.

2.3. Methods

In this study, text analysis of social media data provided keywords by which to code posts into CES: instead of relying on existing CES classification systems and associated keywords [13,17], locally relevant sets were induced by analyzing the posts themselves. The SolVES model is a

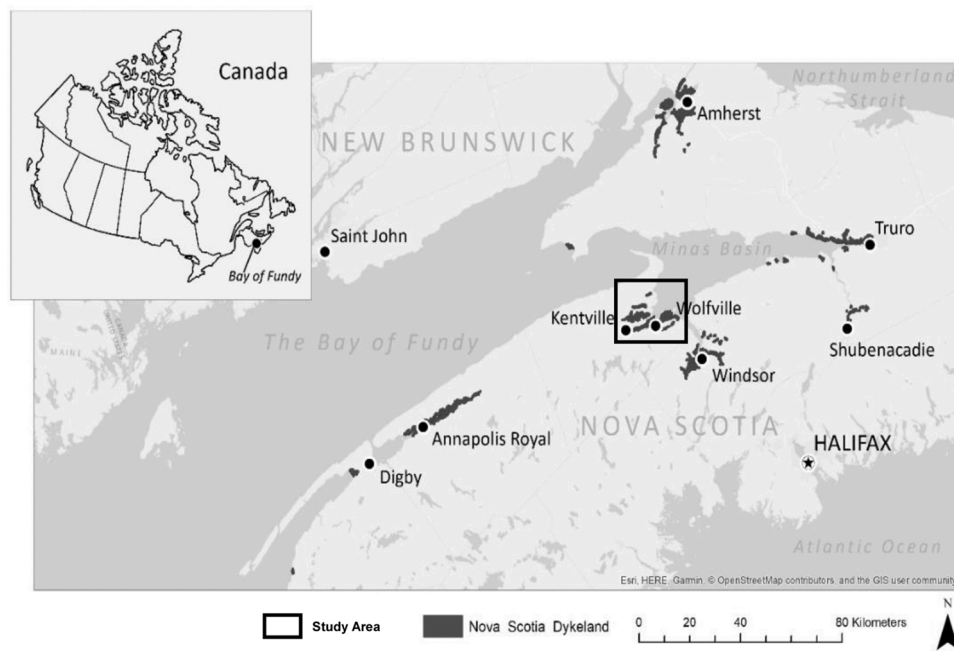


Fig. 1. Study area location.

Table 1
Data Sources.

Data	Data Format	Source
Landform	Raster (30 m)	Canada Centre for Remote Sensing (CCRS)
Elevation	Raster (20 m)	Government of Nova Scotia
LULC	Raster (30 m)	Agriculture and Agri-Food Canada (AAFC)
Population density	Vector	Census of Population
Road	Vector	Government of Nova Scotia
Water bodies (Lakes, Rivers)	Vector	The Nova Scotia Hydrographic Network (NSHN)
Vineyard points	Vector	Manually gathered from Google Maps
Dyke	Vector	6 m buffer around the dyke centreline features from Nova Scotia department of Agriculture (NSDA)
Agricultural dykeland	Vector	Nova Scotia department of Agriculture (NSDA)
Tidal wetland	Vector	Union of Nova Scotia department of Agriculture (NSDA) and The UN Environment Programme World Conservation Monitoring Centre (UNEP-WCMC)
Coast	Vector	The Nova Scotia Hydrographic Network (NSHN)

geospatial tool that enables the evaluation and mapping of multiple CES. Used in conjunction with SolVES, geocoded social media posts classified to CES were combined with environmental data (e.g., the variables hypothesized to drive CES provision), producing a spatially explicit map of cultural services for a given study area. Finally, we examined CES maps to compare CES delivery by our key landscapes, specifically tidal wetlands and agricultural dykelands. Fig. 2 illustrates our work flow.

2.3.1. Identifying CES from text analysis of social media data

The data captured from Instagram included the following information: usernames, posts, web links, number of likes, publication dates, and geographic coordinates. After data collection, processing of the data was performed to prepare and transform the dataset for further analysis more efficiently. The data preprocessing steps included removing user information, duplicates and irrelevant words, segmenting sentences,

and extracting words and sub-words from each post. We then performed text analysis on the processed dataset using the Bigram Language Model [60]. Two-word pairs (i.e. bigrams) are more helpful to understand the meaning within a caption than single words. Bigrams also have the advantages of simplicity and processing speed. Many experiments and studies on social texts and news texts have successfully used bigrams as a research method for statistical language identification [61–63].

After data preprocessing, we extracted 9898 word pairs and their frequency (ranging from 7 to 2020) from the posts. As a team we manually assessed the 395 bi-grams with a frequency higher than 50 to associate them with specific CES where appropriate. We thus induced locally relevant keyword sets by analyzing the posts themselves as well as checking similar keyword lists from other articles to add relevant words to our dataset [13,17,48,64]. 123 final terms or word pairs were identified through bi-gram analysis and helped us to code 17,536 posts to one or more CES categories based on caption contents: aesthetics (10 words or word pairs), cultural heritage/diversity (15), education and knowledge systems (18), inspiration and art (13), recreation and tourism (43), sense of place and terroir (25), social relations and relational values (12) and spiritual (5), all adapted from the categories defined in the Millennium Ecosystem Assessment [21], Table 2). See supplemental material for full details of word pairs; some word pairs are used for more than one CES.

2.3.2. SolVES model

The SolVES model, jointly developed by the USGS Geoscience and Environmental Change Science Center (GECSC), is a tool to comprehensively assess, quantify, and map various cultural services [65]. SolVES derives a quantitative 10-point value index (VI) from a combination of spatial and nonspatial responses to social surveys that assign CES to geographic locations. The Maxent (maximum entropy) modelling software embedded in SolVES can explain the relationship between sites of CES delivery and environmental data, which enables a complete visual mapping of the evaluation results (i.e., how specific CES are statistically associated, if at all, with environmental variables or gradients). After completing the text searches to associate CES keywords to social media data, we selected posts that could be classified as a certain category of CES and used their geographic coordinates (geo-tagged information) to generate points data. The information associated with the CES

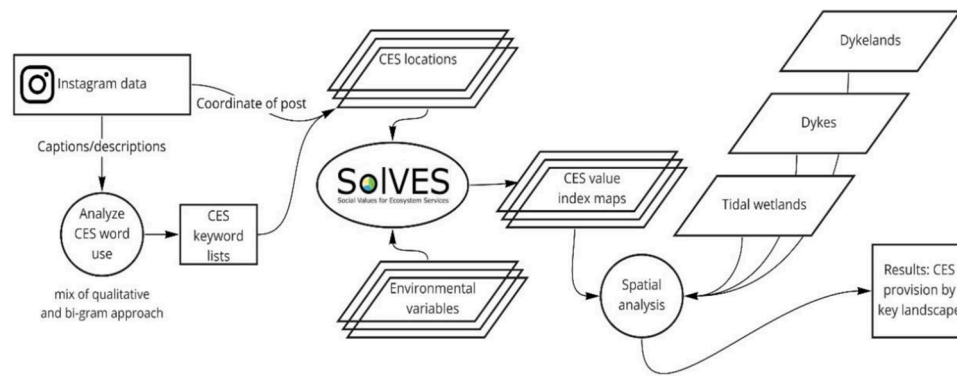


Fig. 2. Work flow chart.

Table 2
Coding structure of landscape CES categories.

CES categories	Description	Proportion of posts coded
Aesthetics	Many people appreciate the beauty or aesthetic value of various aspects of ecosystems, as evidenced by support for parks, scenic drives, and housing location selection.	16.40%
Cultural heritage/diversity	Many societies value the preservation of historically significant landscapes ("cultural landscapes") or culturally significant species.	8.66%
Education and knowledge systems	In many societies, ecosystems and their components and processes serve as the foundation for both formal and informal education, and they influence the types of knowledge systems developed by different cultures.	5.70%
Inspiration and art	Ecosystems are a rich source of inspiration for art (photographs, art structures, etc.), folklore, national symbols, architecture, and advertising.	20.49%
Recreation and tourism	People frequently base their leisure activities on the characteristics of natural or cultivated landscapes in a given area.	35.90%
Sense of place and terroir	Many people value the "sense of place" associated with recognised features of their environment, including aspects of the ecosystem; these features represent a strong identity for people, such as their home or memories associated with a place. Place, including specific soil, topography, climate, landscape characteristics and biodiversity features, and human factors contribute to terroir.	52.43%
Social relations and relational values	Ecosystems have an impact on the types of social relationships that emerge in different cultures. Fishing societies, for example, differ in many ways from nomadic herding or agricultural societies in terms of social relations.	13.65%
Spiritual	Many religions attribute spiritual and religious significance to ecosystems or their components.	4.47%

categories were stored together in the model's geodatabase in the form of table data. These data were imported into SolVES instead of survey data to evaluate and map the CES. Since the coordinates of most of the posts were in developed areas, we resampled the high-density areas for the purpose of a flatter distribution of posts (see [supplemental material](#) for full details of the sampling). After resampling, 8776 posts were entered into modeling in the study area.

2.3.3. Analysis of delivery CES patterns in each landscape

The key landscapes analyzed in this study include dykes (58 ha), dykelands (3659 ha), and tidal wetlands (1141 ha). The dyke structures themselves (modelled as a 6 m buffer zone on each side of the dyke lines) can serve as an important carrier for the delivery of services because people travel along them to access the coast and recreate. Therefore, they were included in the analysis, but the locational accuracy of the geo-location of the posts was unlikely to capture such narrow features well. The spatial distribution of the key landscapes is shown in [Fig. 3](#). According to the spatial location of services as output from SolVES, we analyzed the CES information contained in each key landscape, including: the total value of CES delivered by each landscape, the value of CES delivered per ha area of each landscape, and—following reclassification (0–100) to remove low-value areas (VI from 0 to 32), given the potential for social media noise or categorization error—which key landscapes provide the most CES value area. At the same time, we overlaid the layers of each CES, thus showing the spatial divergence of the number of CES provided by the region. In so doing, the cultural services delivery patterns of each key landscape can be clearly compared and communicated to people.

3. Results

3.1. Social media data description

The captions of Instagram posts provided a rich source of information about CES. The number of posts we used to model each CES ranged from 392 (spiritual) to 4601 (sense of place and terroir), and were posted by between 161 and 1922 users, respectively. These two CES also had the highest average number of posts per poster included in the study (2.4). The lowest number of posts per poster was 1.6 (social relations and relational values) (see [supplemental material](#) for full details of user statistics). Due to concerns about individual posters having an outside influence on our corpus, we also looked at outliers in the dataset. We set a threshold to explore how many users made a relatively large contribution to the service, as defined by 2 standard deviations above the mean (see [supplemental material](#) for full details of user statistics). The two services that had the most outliers were sense of place and terroir (39 outliers), and recreation and tourism (37), but the highest individual posters still only comprised 2.4% and 2% of the posts in those CES categories, respectively. One person (a visual artist) above the threshold in the spiritual CES category comprised almost half of the posts in that category (46.2%); to account for this, we included the posts in our modelling but placed less weight on them in the results. Similarly, a business owner produced 10.8% of the posts in the education and knowledge systems category. In all other categories the influence of impactful individuals was between 2% and 5%.

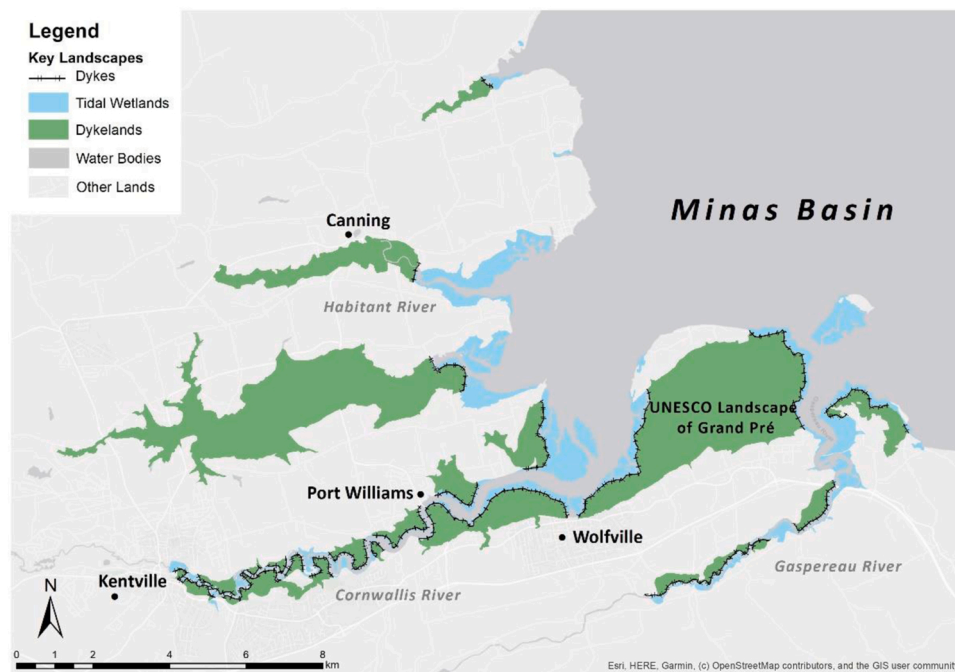


Fig. 3. Key landscapes in the study area: dykes, dykelands, and tidal wetlands.

3.2. CES modelling and multifunctionality

Value index (VI) maps that highlight the medium and high indices show that the spatial distribution of CES in the study area reasonably overlap across categories (Fig. 4). Our SolVES models demonstrated that services were highly clustered in developed areas and near transportation, even after our attempts to reduce the density of posts in those areas, compared with the more peripheral dykelands and wetlands of interest. CES were also associated with water bodies, coast, and distances to different key landscapes: as distance increases, CES values gradually decrease. The value of each service also tended to increase with population density, although four CES (cultural heritage/diversity, sense of place and terroir, recreation and tourism, and aesthetics services) also had med-to-high values in areas with lower population density (see [supplemental material](#) for full details of environmental metrics). In general, CES were largely driven by demographic and transport factors.

Overlaying all services demonstrates the multifunctionality of the landscape; the red colours are where all 8 CES were modelled to occur (Fig. 5). The areas with a strong ability to provide multiple services in the overlap map were concentrated in developed areas such as Wolfville and Grand-Pré. Kentville, Port Williams, and Canning areas were also multifunctional in CES terms.

3.3. CES delivery patterns in key landscapes

Generally, the posting activity in developed areas makes it difficult to see what the key landscapes themselves foster. In all the key landscapes service delivery by area was low (Table 3). In fact, 95–99% of all key landscape areas were modelled to have no CES supply, while 1–2.5% were modelled to have low CES (bottom third of value index).

Understanding the value of dykelands, dykes and tidal wetlands in CES terms using social media modelling requires us to recalibrate, rather than compare them to high-visitation amenity areas such as towns and vineyards. Here we focus on how VI metrics were distributed in our key landscapes (Table 4). Dykelands delivered more value for all CES because they cover a larger area. On a per-unit basis, however, wetlands provided more CES in all categories except spiritual (where dykes

provided the most) and cultural heritage/diversity (where dykelands provided the most). Adding all CES, the same pattern holds: more area of CES value is provided by dykelands, but more CES per unit area are provided by wetlands.

Inevitable social media noise and errors introduced by the keyword approach led us to focus more on areas modelled to have med-high CES value indices (top two thirds), to avoid overestimation. Filtering out low values, which dominated the dataset, the area of modelled dykeland CES was higher for three CES: cultural heritage/diversity, education and knowledge systems, social relations and relational values (Table 5). Wetlands had a higher modelled area in sense of place and terroir. Wetlands and dykelands were similar in terms of aesthetics, inspiration and art, and recreation and tourism. In general, tidal wetlands were more multifunctional, with an average of 2.6 services provided per ha of dykelands, where med-high levels of services exist, compared with 1.4 services for tidal wetlands, and 0.3 services provided by dykes. The dyke structures themselves were small and hard to understand using these kinds of metrics. While they did not offer a higher ability to deliver any CES, they were likely transportation routes for delivering many.

4. Discussion

Making informed coastal adaptation decisions in the rural Bay of Fundy study area, as the Nova Scotia Department of Agriculture is currently doing, requires an understanding of how alternative landscape options might affect nearby residents and other landscape users. Previous research has struggled to elicit the values and ecosystem services related to tidal wetlands, existing as they do at the intersection of land and ocean. Our research approach using Instagram data chained text analysis, ecosystem service modelling and spatial analysis tools to show that ES delivery differs between dykelands and tidal wetlands. It also illustrated the ongoing challenges of doing such work in rural and peripheral areas. This discussion highlights some of our key findings and their implications, as well as limitations of the approach and next steps for future research.

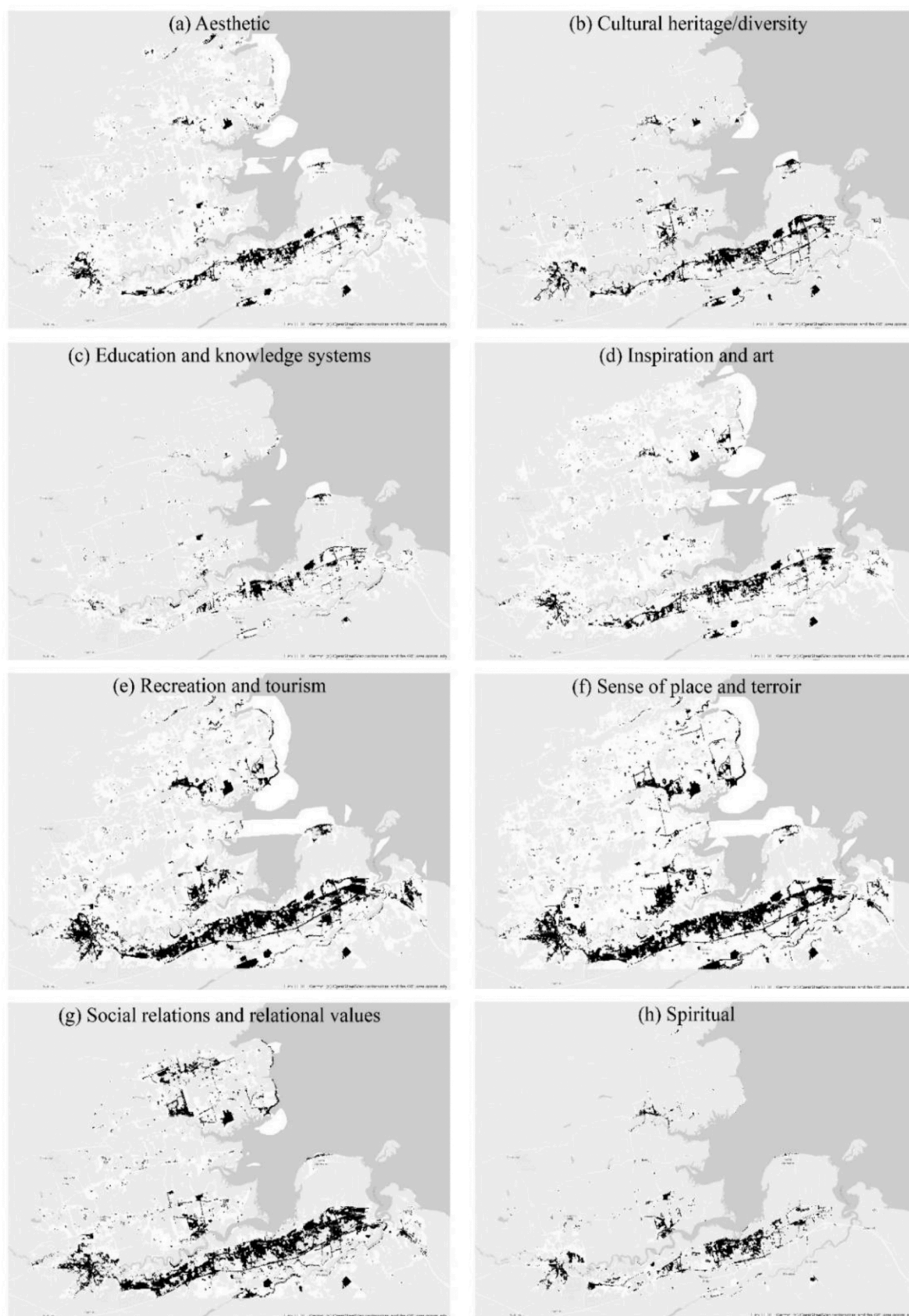


Fig. 4. The VI map of CES distribution, (a) Aesthetics, (b) Cultural heritage/diversity, (c) Education and knowledge systems, (d) Inspiration and art, (e) Recreation and tourism, (f) Sense of place and terroir, (g) Social relations and relational values, (h) Spiritual. We divide the values into two categories: none/low (0) and med to high (1), with 0 in white and 1 in black. Water is in grey.

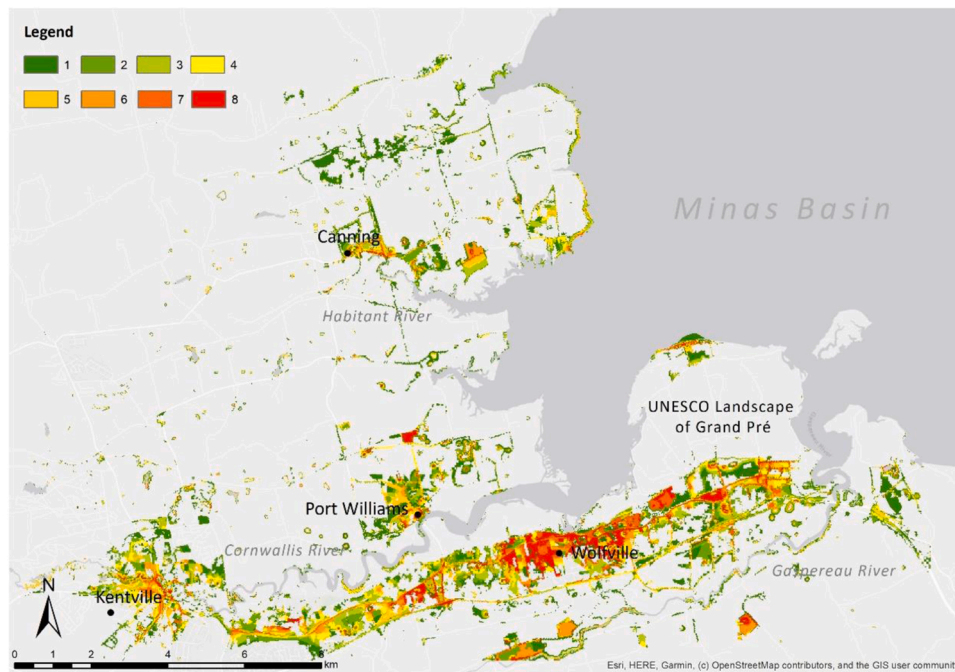


Fig. 5. The overlay VI map of CES distribution.

Table 3

The area of VI in each landscape and the ratio of the total area.

Value Index	Dykes (ha)	Dyklands (ha)	Wetlands (ha)	Dykes % area	Dyklands % area	Wetlands % area
No value	57.20	3578.96	1086.32	98.62%	97.81%	95.21%
Low	0.76	31.64	28.08	1.31%	0.86%	2.46%
Med-high	0.04	48.40	26.60	0.07%	1.32%	2.33%

Table 4

CES of each landscape delivery as estimated by summing VI within key landscape areas, and per ha only where services are modelled as present.

CES categories	All CES delivery (summed VI in cells)			CES delivery (VI per ha)		
	Dykes	Wetlands	Dyklands	Dykes	Wetlands	Dyklands
Aesthetics	59	7961	18,675	1.02	6.98	5.25
Cultural heritage/diversity	16	1433	7222	0.28	1.26	2.03
Education and knowledge systems	18	2833	8531	0.31	2.48	2.40
Inspiration and art	150	6405	13,056	2.59	5.61	3.67
Recreation and tourism	446	15,910	37,151	7.69	13.94	10.44
Sense of place and terroir	815	20,761	48,738	14.05	18.20	13.70
Social relations and relational values	74	5064	12,021	1.28	4.44	3.38
Spiritual	93	558	4778	1.60	0.49	1.34
All values	1671	60,925	150,172	28.81	53.39	42.22

Table 5

Area in hectares of med-high value areas by CES category and landscape*.

CES categories	Dykes (ha)	Wetlands (ha)	Dyklands (ha)
Aesthetics	0.00	10.92	9.00
Cultural heritage/diversity	0.00	3.80	30.72
Education and knowledge systems	0.04	8.68	22.56
Inspiration and art	0.00	7.00	6.68
Recreation and tourism	0.20	35.12	39.40
Sense of place and terroir	0.20	31.60	22.00
Social relations and relational values	0.08	19.60	44.24
Spiritual	0.24	2.12	36.64
Average number of CES per ha*	0.33	1.44	2.60

* Only where medium or high CES value exist; not zero or low.

4.1. Pattern analysis of CES delivery in different landscapes

In our analysis, both dykelands and wetlands had low overall CES coverage. Where services were modelled as present, some CES delivery capacity was comparable across landscapes. For other CES, the landscapes had very different delivery capabilities. Overall, dykelands provided more services in our ES modelling, likely because of their larger area, but perhaps also because their inland position placed them in closer proximity to developed areas with amenities that were dominant in our dataset. A previous study using manual coding of the same dataset reported strong associations of dykes and dykelands with aesthetics, recreational use, and social relations [17], and another using Q-method confirmed the cultural and recreational values of dykelands [52]. This study reaffirms that dykelands provide these cultural and recreational services, and more besides.

Despite their more peripheral position, wetlands showed a stronger ability in our analysis to provide multiple services per unit area, consistent with their status as ecosystem service ‘superstores’ [66]. Wetlands showed stronger delivery in the combined sense of place and terroir category. This is likely due to the impact of sense of place, given that local food and drink are associated with terroir and would logically be more present in agricultural dykelands. In terms of delivery of aesthetics, inspiration and art, recreation and tourism CES, dykelands and wetlands are comparable. While peripheral relative to population concentrations, wetlands sit between the dykes and ocean, and the former are often used for water access and photographing ocean views. Without close examination of the photos, it is difficult to discern whether the wetlands or ocean sparks these values, especially since people capturing tidal wetlands in social media posts do not tend to call them by name [17]. This demonstrates a lack of local familiarity with the ecosystem, perhaps because of local extirpation, estimated to be 85% since European settlement [18]. Despite landscape differences, the top three services delivered by dykelands and wetlands are the same: sense of place and terroir, recreation and tourism, and aesthetics. This may be because people interacting with dykelands, dykes and wetlands view them as an inseparable complex: they are landscape elements that have been found together in this region for hundreds of years and may seem to belong together. Thus, while we have been seeking to understand trade-offs between landscapes in this work, it is likely that the loss of any of the elements has costs for users.

Most of the key landscape areas presented as having no CES value, but that does not mean they have no value or capacity for CES provision. It is more likely that we cannot measure the values using this dataset and approach. It is important to recognize that you don’t always need to go inside a landscape to benefit from its corresponding services. This is especially true for the low-lying landscape of dykelands and tidal wetlands. A good example comes from the historic Grand-Pré dyke, where the best views are not from the dyke, but the upland areas behind it. A photo captured there would not register as a service delivered by the dykeland, although it likely is, and a hotspot from multiple such photos would be misinterpreted. Viewshed analysis may be a better way to proceed with such studies [24,67]; capturing not where photos are taken but the area the photos capture.

4.2. Informing coastal infrastructure

Looking at the study area through a climate adaptation lens, it is important to note that without human intervention, many of the current dykes in the Bay of Fundy would have been flooded by rising sea levels and storms, naturally converting to tidal wetlands [68]. Looking at the ecosystem services framework, we know that sacrificing dykelands to expand tidal wetlands in carefully chosen sites can provide sustainable protective services compared to dykes [18]. In the context of the study area, the primary adaptation approach being discussed by the Department of Agriculture is dyke realignment, not wholesale dyke/dykeland removal or raising (with wider footings, further squeezing foreshore marshes). Dyke realignment involves pulling inland and often straightening dyke infrastructure to make room for tidal wetland restoration, maintaining the abovementioned complex, compared to all-or-nothing solutions [3]. Our analysis suggests that there is unlikely to be a significant cost in CES from dyke realignment, unless access itself is lost (via the dykes or other comparable trail infrastructure). Landscape and place are the connecting links between ecosystem function outcomes and cultural services, so it is necessary to preserve the most iconic sociocultural landscape features [69]. Restoring wetlands in current low-value dykeland areas where there are few overall delivery services and taking reinforcement measures in med-high dykeland value areas, may be a feasible option for maintaining existing services to the greatest extent. This approach preserves dykelands’ vital services while increasing services from tidal wetlands, providing a balanced option for retaining the necessary CES [70].

Finally, ES modelling demonstrated the same bias toward transportation and population density that has been seen in other work [22, 71], with the notable exception of the dykes themselves. Dyke structures enable people to carry out activities and obtain services in both dykelands and wetlands, providing the elevated position from which many of the photos in those landscapes may have been captured. However, because of their narrowness, it is difficult to determine their significance using our methods. Nevertheless, they remain likely to be important routes for recreation or access points for activities connected to the other landscapes [72]. Alternative data sources such as those used for fitness tracking (e.g. Strava) or naturalist activities (e.g. eBird or iNaturalist) might help balance the biases of development in Instagram-based datasets, and target more specific services such as birdwatching [73].

4.3. Limitations of a social media approach

Social media data has been increasingly used as a data source for CES evaluation in recent years, but it still has some limitations. Due to issues such as user privacy and changes in data access policies [74,75], social media data has become difficult to obtain, weakening the capacity for replication and longitudinal research. For example, it is no longer possible to obtain Instagram data through Netlytic as we did, nor is it easy to retrieve comparable data through other approaches (e.g., web scraping tools). Use patterns among demographics also shift over time, affecting longitudinal work. The high proportion of young users on the platform may affect the representativeness of the data overall. Additionally, some individuals have highly variable posting behaviour, which can lead to biased data. For example, we disregarded the spiritual CES results here because half (46%) of the 392 posts came from one person. Various approaches to filtering or sampling can be done to reduce such effects from over-active users [76,77]. Additionally, using a word-based approach to identify CES can be plagued by multiple and playful meanings of words (e.g., puns), especially when the posts were not created for research purposes. Biases also came from different types of users on the platform (including marketers), and geographically from mis-matched locations where posts were geo-tagged but did not present the photo contents. Fine-grained key landscapes such as ours are more vulnerable to such errors. These limitations are thought to decrease with the continuing increase in number and awareness of social media users and in data accuracy allowed by newer technologies and the combination of different data sources [78,79]. Social media data can better provide a reliable representation of CES when researchers include image content and enrich the data with a measure of sentiment [80]. Nevertheless, as an emerging data source, social media data provides a wealth of information for studying CES, and triangulation with previous studies suggests the validity of the results.

5. Conclusions

Attention to coastal risk issues caused by climate change and sea level rise has become the focus of international discussions. After centuries of development, the Bay of Fundy dyke system has shaped agricultural dykelands and transformed tidal wetlands, impacting the people connected to them through their provision of ecosystem services. Increasingly diverse uses of dykelands, dykes, and tidal wetlands play a role in the delivery of ES, and more specifically CES. This study focused on the delivery patterns of tidal wetlands and agricultural dykelands across different CES, mapping the spatial distribution and multifunctional areas of various CES in the study area. This approach provided trade-off information for those working to cope with regional climate change and spatial insight for identifying specific dykes to adapt.

At the same time, the combination of social media data and other modeling methods was explored to provide new research ideas for the utilization of social media data. The results show that agricultural dykelands have a higher supply capacity for cultural heritage/diversity, education and knowledge systems, social relations and relational values.

Tidal wetlands have a higher supply capacity for sense of place and terroir. In terms of aesthetics, inspiration and art, and recreation and tourism, the two landscapes are quite comparable, and our research indicates that the landscapes may combine to form a valued complex. Both agricultural dykelands and tidal wetlands show multifunctionality where CES are present, with an average of 1.4 services provided by a given area of agricultural dykeland and 2.6 services provided by tidal wetlands. In the future, these methods will need to be triangulated with other methods. Triangulation should include comparisons with survey-based methods to yield better insights from different data sources. Applying similar methods to other social media or media sharing platforms would also help with data validation.

CRedit authorship contribution statement

Qiqi Zhao: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Yan Chen:** Methodology, Investigation, Writing – review & editing. **Keshava Pallavi Gone:** Data curation, Writing – review & editing. **Emily Wells:** Methodology. **Keahna Margeson:** Methodology, Writing – review & editing. **Kate Sherren:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Visualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors are unable or have chosen not to specify which data has been used.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.marpol.2023.105533](https://doi.org/10.1016/j.marpol.2023.105533).

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